

AI/ML Internship – Day 54 Notes & Tasks

Module 7: Bag of Words (BoW) – Converting Text into Numerical Features for Machine Learning

Part 1: Recap of Day 53

Yesterday we learned:

-  Stemming
-  Lemmatization
-  Root Words
-  Vocabulary Reduction

Today we will learn:

Bag of Words (BoW)

One of the most important concepts in NLP and Machine Learning.

Part 2: Why Do We Need Bag of Words?

Computers cannot understand text directly.

Example:

I love AI

Humans understand the meaning.

Computers only understand:

Numbers

So we need to convert text into numbers.

This process is called:

Feature Extraction

Part 3: What is Bag of Words?

Definition

Bag of Words (BoW) is a technique used to convert text into numerical vectors based on word frequency.

Simple Idea

Instead of understanding meaning, BoW counts how many times each word appears.

Part 4: Example Dataset

Sentence 1:

I love AI

Sentence 2:

I love Python

Part 5: Create Vocabulary

Find all unique words:

I

love

AI

Python

Vocabulary:

[I, love, AI, Python]

Part 6: Convert Sentence to Numbers

Sentence:

I love AI

Count each vocabulary word:

Word	Count
------	-------

I	1
---	---

love	1
------	---

Word Count

AI 1

Python 0

Vector:

[1,1,1,0]

Part 7: Second Sentence

Sentence:

I love Python

Vector:

Word Count

I 1

love 1

AI 0

Python 1

Result:

[1,1,0,1]

Part 8: Visual Representation

Vocabulary:

[I, love, AI, Python]

Sentence:

I love AI

↓

[1,1,1,0]

Sentence:

I love Python

↓

[1,1,0,1]

Part 9: Why is it Called "Bag" of Words?

Imagine a bag containing words.

BoW cares about:

- ☒ Which words exist
 - ☒ How many times they appear
-

BoW ignores:

- ☒ Grammar
 - ☒ Word order
 - ☒ Sentence structure
-

Example:

I love AI

and

AI love I

produce almost the same representation.

Part 10: Larger Example

Sentence:

AI is amazing and AI is powerful

Vocabulary:

AI

is

amazing

and
powerful

Count words:

Word	Count
AI	2
is	2
amazing	1
and	1
powerful	1

Vector:

[2,2,1,1,1]

Part 11: BoW Workflow

Raw Text



Cleaning



Tokenization



Stop Word Removal



Vocabulary Creation



Word Counting



Numerical Vector

Part 12: Why Remove Stop Words Before BoW?

Example:

The cat is sitting on the chair

Without removing stop words:

the
is
on

appear frequently.

These words add little meaning.

After stop word removal:

cat
sitting
chair

More useful information.

Part 13: BoW Using Python

```
from sklearn.feature_extraction.text import CountVectorizer
```

```
documents = [  
    "I love AI",  
    "I love Python"  
]
```

```
vectorizer = CountVectorizer()
```

```
X = vectorizer.fit_transform(documents)
```

```
print(X.toarray())
```

Output:

```
[[1 1 1 0]  
 [1 1 0 1]]
```

Part 14: View Vocabulary

```
print(  
    vectorizer.get_feature_names_out()  
)
```

Output:

```
['ai' 'love' 'python']
```

Part 15: Understanding Matrix Output

Example:

```
[[1 1 1 0]  
 [1 1 0 1]]
```

Meaning:

Sentence 1:

I love AI

↓

```
[1 1 1 0]
```

Sentence 2:

I love Python

↓

```
[1 1 0 1]
```

Part 16: Real-World Example

Movie Reviews

Review 1:

Amazing movie

Review 2:

Bad movie

Vocabulary:

Amazing
Bad
Movie

Vectors:

Amazing movie

[1,0,1]

Bad movie

[0,1,1]

Now machine learning algorithms can process them.

Part 17: Applications of BoW

Sentiment Analysis

Positive vs Negative Reviews

Spam Detection

Spam vs Not Spam

Chatbots

Understanding text input

News Classification

Sports

Politics

Technology

Email Categorization

Different email categories

Part 18: Advantages of BoW

- ☒ Easy to understand
 - ☒ Easy to implement
 - ☒ Fast
 - ☒ Good for beginners
-

Part 19: Limitations of BoW

Ignores Word Order

Example:

I love AI

and

AI love I

look similar.

Ignores Meaning

Example:

Good

and

Excellent

are treated as different words.

Large Vocabulary Problem

Large datasets create huge vectors.

Part 20: Theory Questions

What is Bag of Words?

A technique that converts text into numerical vectors using word frequencies.

Why do we need BoW?

Computers understand numbers, not text.

What is a Vocabulary?

Collection of unique words in a dataset.

Does BoW consider word order?

No.

What does BoW count?

Word occurrences.

Part 21: Interview Questions

Explain Bag of Words.

BoW converts text into numerical features by counting word frequencies.

What are the limitations of BoW?

Ignores word order and meaning.

Why is BoW important?

It allows machine learning models to process text data.

What is CountVectorizer?

A Scikit-Learn tool used to create Bag of Words representations.

What is Feature Extraction?

Converting raw data into numerical features for machine learning.

Tasks for Day 54

Theory

1. What is Bag of Words?
 2. Why do we need BoW?
 3. What is a vocabulary?
 4. Explain CountVectorizer.
 5. List BoW limitations.
-

Practical Task 1

Create vocabulary from:

AI is powerful

AI is amazing

Practical Task 2

Convert:

I love AI

into a BoW vector manually.

Practical Task 3

Install Scikit-Learn:

```
pip install scikit-learn
```

Practical Task 4

Implement CountVectorizer.

Practical Task 5

Generate BoW vectors for 5 sentences.